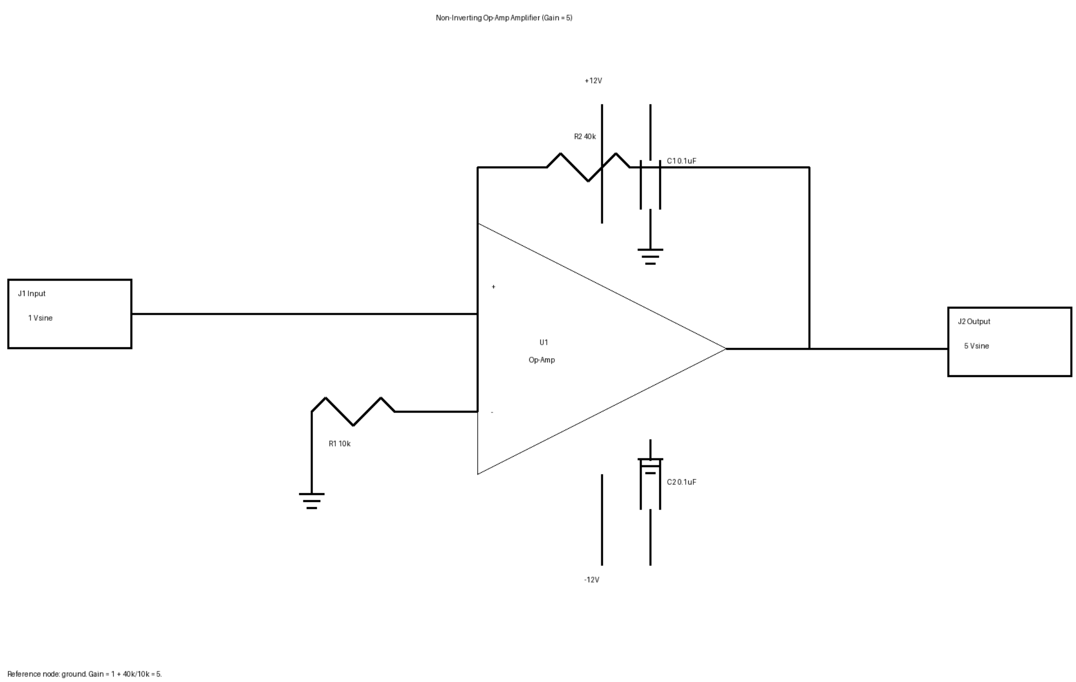
Op-Amp Signal Amplifier (1 V sine to 5 V sine)

Goal: amplify a 1 V sine-wave input to approximately 5 V output amplitude with a gain-of-5 non-inverting op-amp stage.

Assumptions

- Input is a 1 V amplitude sine wave centered around ground.
- Target output is a 5 V amplitude sine wave, requiring a voltage gain of about 5.
- A dual-rail supply is assumed for clean ground-centered sine amplification.
- Chosen topology is a non-inverting op-amp with gain 1 + Rf/Rg = 5.
- Example supply rails are +12 V and -12 V.
- Circuit is intended as a conceptual low-frequency analog amplifier example.

Schematic



Bill of Materials

RefDes	Description	Qty	Unit $	Ext. $	Supplier / Link
U1	General-purpose op-amp, e.g. TL081/TL071	1	0.80	0.80	TL081CP https://www.digikey.com/en/products/filter/linear-amplifiers-instrumentation-op-amps-buffer-amps/687
R1	10 kohm resistor, feedback-to-ground gain resistor	1	0.05	0.05	Generic https://www.digikey.com/en/products/filter/through-hole-resistors/53

R2	40 kohm resistor, feedback resistor	1	0.05	0.05	Generic https://www.digikey.com/en/products/filter/through-hole-resistors/53
C1	0.1 uF ceramic bypass capacitor for +V rail	1	0.05	0.05	Generic https://www.digikey.com/en/products/filter/ceramic-capacitors/60
C2	0.1 uF ceramic bypass capacitor for -V rail	1	0.05	0.05	Generic https://www.digikey.com/en/products/filter/ceramic-capacitors/60
J1	Input connector / signal source interface	1	0.50	0.50	Generic https://www.amazon.com/s?k=terminal+block+connector
J2	Output connector / load interface	1	0.50	0.50	Generic https://www.amazon.com/s?k=terminal+block+connector
PS1	Dual supply source, +/-12 V bench supply or equivalent	1	15.00	15.00	Illustrative https://www.amazon.com/s?k=dual+power+supply+bench
	Total estimated cost			17.00	

No firmware is required because this is a purely analog amplifier.

Design note: if the input signal is not ground-centered or if only a single supply is available, biasing and common-mode considerations must be revisited.